AMES HOUSING DATA ANALYSIS REPORT

INTRODUCTION

For this project, I used the Ames Housing dataset, which contains housing records with a wide range of numerical and categorical variables related to property size, quality, condition, and sale details. I selected this dataset because it offers a realistic example of messy real-world data and provides enough variety to make cleaning, feature engineering, and analysis meaningful.

The main goal of this project was to understand which factors appear to have the strongest relationship with house prices and to build a complete analysis pipeline starting from raw data. More specifically, I wanted to examine the main issues in the dataset, improve the data through feature engineering, and identify useful patterns that could explain variation in SalePrice.

CLEANING SUMMARY

Before starting the analysis, I cleaned the raw dataset to make sure it was ready to use. I first looked at the shape of the data, checked the data types, and found the missing values so I could understand the main issues. After that, I put all the cleaning steps into a reusable clean_data() function to keep the work organised across the notebooks.

The cleaning mainly focused on missing values, data type issues, duplicate checks, and extreme values in the target column. I handled each problem in a way that fit the column instead of using one method for everything. For SalePrice, I capped the highest values at the 99th percentile so they would not heavily affect the analysis, while still keeping the records in the dataset. I also added a few checks at the end to make sure the cleaned data was ready for the next steps.

FEATURE SUMMARY

After cleaning the data, I created and transformed several features to make the dataset more useful for analysis. I encoded categorical columns such as House Style and Sale Condition, converted Heating QC into ordered numeric values, and scaled Lot Area and Gr Liv Area. I also created features such as area_per_room, baths_per_bedroom, and house_value_score, then applied a log transform to Lot Area and grouped house ages into categories to make the data easier to interpret.

KEY FINDINGS

Finding 1: Overall quality had the strongest relationship with sale price

One of the clearest results in the analysis was that Overall Qual showed the strongest correlation with SalePrice (r = 0.81). The heatmap also showed that features such as Gr Liv Area, Garage Cars, and Garage Area were strongly related to price. This suggests that both house quality and size-related features play an important role in explaining house prices.

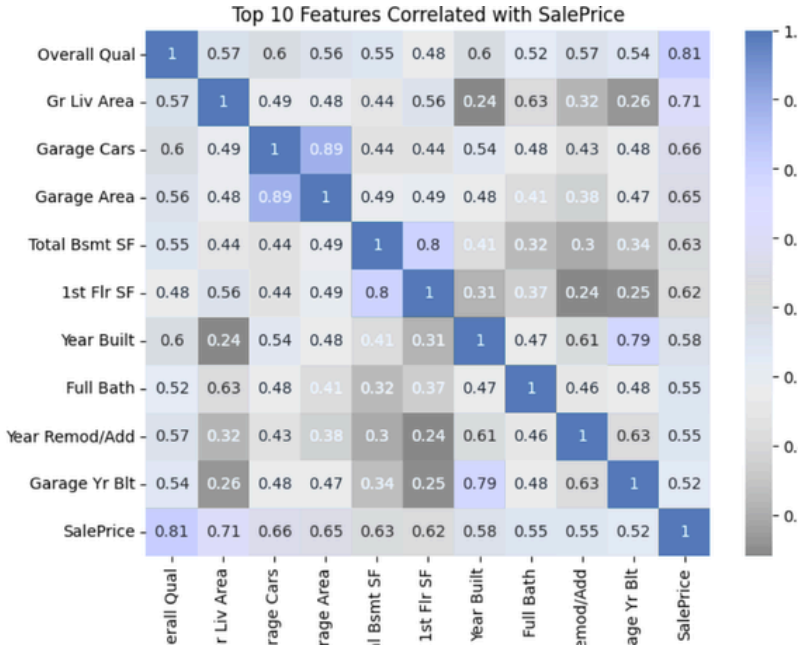

Figure 1. Heatmap showing the top 10 numerical features most correlated with SalePrice.

Finding 2: Some numerical features needed transformation

The raw data was not evenly distributed, especially in features related to price and area. Some columns were clearly skewed and included a few extreme values, which could affect the analysis if left untreated. This is why steps such as capping outliers and applying transformations were important before moving on to the visual and statistical analysis.

Finding 3: Engineered features made the data easier to interpret

The engineered features made the dataset easier to understand than the original columns alone. For example, area_per_room gave a clearer idea of how space is distributed inside the house, while house_value_score combined size and quality in one variable. These features helped make the results more meaningful and easier to explain.

What I Would Do Next

If I had more time, I would continue this project by building a simple model to predict SalePrice and compare the results before and after feature engineering. This would help show whether the new features improved the dataset in a practical way, not only in interpretation but also in prediction.

I would also explore feature importance in more detail to identify which variables have the strongest effect on house prices. Another useful next step would be to create a small dashboard that brings the main charts together in one place and gives a clearer overall view of the dataset.